

CRIMECAST

Crime Analysis, Prediction & Sentiment Analysis
Using Machine Learning (Tamil Nadu Districts)

VIVA PREPARATION GUIDE

How the project works · Expected examiner questions

College prototype · Not a live police system · Not an official SCRB forecast

1. Elevator pitch (memorise)

“CRIMECAST cleans multi-table Tamil Nadu crime stats, trains classical ML for district rates and counts with honest temporal metrics, adds news sentiment for live context, and shows 2026 as a **scenario** with maps, compare, and explainability — not as official SCRB output.”

2. How it works — block by block

Pipeline: **Data** → **Clean** → **Train** → **Predict** → **News/Sentiment** → **Dashboard**, with a parallel **2026 trend scenario** branch.

Block 1 — Data (inputs)

Source	Role
Official-style CSVs (complaints, murder, women crimes)	Numbers for ML; train mainly on official-era years ≤ 2023
News harvest (Tamil + English)	Live heat, sentiment, word clouds — not FIRs
SQLite (data/crimecast.db)	Headlines, alerts, optional migrated tables

Say: We separate official labels used for training from media used for live awareness.

Block 2 — Clean (clean_data.py)

- Discover yearly CSVs; fix headers (e.g. muder → murder)
- Drop TOTAL aggregate rows (no double counting)
- Merge into crimecast_ml_ready.csv (district × year)
- Map cities to 38 TN districts (Madurai City→Madurai, Avadi→Chennai)

Say: One ML-ready table, consistent districts.

Block 3 — Train (train_model.py)

- Targets: complaints, murder count/rate, rape count/rate, cognizable rate
- Models: Dummy, Ridge, Random Forest, Gradient Boosting (often log targets)
- Select best via CV + temporal holdout where possible
- Save models/*.joblib + training_metrics.csv

Say: We don't claim perfect forecasting; we show temporal metrics honestly.

Block 4 — Predict (predict.py)

- Resolve district (TN38) → build features for year
- Model raw prediction → blend with district history
- Rates lean more on history so rankings stay realistic
- Optional: populate all 38 districts; map has no news fill

Say: Blend keeps sticky rates (e.g. high murder-rate districts) from going wild.

Block 5 — 2026 scenario (forecast_engine)

- Per-district trend on history (not Prophet / LSTM)
- Methods: linear · last year · blend
- Uncertainty bands (low–mid–high)
- Scenario only — not SCRB official

Say: Short time series → simple trend + bands, more honest than black-box deep models.

Block 6 — News + sentiment

- Harvest headlines → DistilBERT if available, else lexicon
- District concern score + word clouds
- Live Feed = news volume in a time window

Say: Media tone and volume, not crime registration.

Block 7 — Dashboard (demo UI)

Tab	One line
Live Feed	News heat + alerts
District Map	Choropleth / heat / per-lakh scoreboard
Accuracy	Metrics, claim/don't claim, blend check
Predict	One district + all-district map
Sentiment	Map + word cloud
2026	Multi-target scenario map + bands
District Compare	Side-by-side + carve maps + brief
Risk Explain	Why high risk (composite + LIME-style)
Health	Files/models ready?

3. Five-minute demo path

1. **Live** — “This is news heat, not FIRs.”
2. **Map** — official rates / density; per-lakh ranking.
3. **Accuracy** — test R^2 , temporal limits, what we claim.
4. **Predict** — one district + drivers.
5. **2026** — scenario + uncertainty.
6. **Compare** — e.g. Thoothukudi vs Madurai.
7. **Explain** — risk factors.

4. Expected examiner questions

A. Problem & scope

Question	Short answer
What is CRIMECAST?	TN district crime analysis + ML prediction + news sentiment + Streamlit demo.
Is this a police system?	No — academic prototype only.
Out of scope?	Real-time FIR system, causal policy proof, official 2026 SCRB authority.

B. Data

Question	Short answer
Data sources?	SCRB/NCRB-style tables + news harvest.
Why remove TOTAL rows?	Avoid double counting state aggregates.
What is TN38?	38 official districts; cities rolled into parents.
News = crime?	No. Volume/tone support only.
Training years?	Mainly official-era ≤ 2023 ; later years thinner/proxy for maps.

C. Machine learning

Question	Short answer
Which algorithms?	Ridge, RF, GB (+ dummy); best per target.
Evaluation?	CV MAE/RMSE/ R^2 + temporal holdout.
Why weak temporal R^2 sometimes?	Few official years $\rightarrow$ hard year-to-year generalisation (we show it).
What is blend?	Model + district history; rates weight history more.
Why not LSTM/Prophet?	Short yearly series; simple models more honest.
Leakage control?	Exclude target's own source family from features.

D. 2026 forecast

Question	Short answer
How is 2026 made?	District trend: linear / last-year / blend + bands.
Official SCRB?	No — scenario for discussion.
Uncertainty?	Residual/horizon-style low–mid–high band.

E. Sentiment & explainability

Question	Short answer
DistilBERT role?	Headline polarity when available; lexicon fallback.
Word cloud?	Frequent terms in district headlines.
SHAP/LIME?	Importance×z or local linear (LIME-style); optional real SHAP.

F. Engineering & testing

Question	Short answer
Why Streamlit?	Fast interactive maps/tables for demo.
Why SQLite?	Zero-server local store; CSVs remain backup.
Postgres?	Future only — not full production path yet.
Testing?	Unit + integration + data quality + UI checklist; <code>py -3 run_tests.py</code> .
No news fill on predict/2026 maps?	Colours mean model/forecast, not media volume.

G. Ethics & limits

Question	Short answer
Bias?	Under-reporting, media bias, uneven district reporting.
Misuse?	Must not drive arrests/policing automatically.
Limitations?	Short series; media ≠ crime; scenario forecasts.
Future work?	More official years, monthly data, multi-user DB, deeper Tamil NLP.

5. If stuck — safe lines

- “That’s a scenario, not an official forecast.”
- “That’s news volume, not registered crime.”
- “With only a few official years we prefer simple, interpretable methods.”
- “We blend with history so rate rankings stay realistic.”
- “We report temporal metrics honestly — even when they are weak.”

6. Day-of checklist

- Restart Streamlit; open `http://localhost:8501`
- Walk demo path (Section 3) in ~5 minutes
- Know one Accuracy number if asked (e.g. temporal R^2 for a rate target)
- Know unit vs integration vs UI testing (`run_tests.py` + checklist)
- Stay calm on limitations — honesty scores well

7. Quick commands

Dashboard: `py -3 -m streamlit run dashboard.py --server.port 8501`

Tests: `py -3 run_tests.py`

Health: `py -3 health_check.py`

— End of viva prep — · CRIMECAST academic prototype